Thermophysical Properties and Petrology of the NWA 11641 Ureilite Meteorite

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Abstract: **Using a Quantum Design Physical Property Measurement System, we have measured the thermal conductivity and heat capacity of the NWA 11641 ureilite meteorite at low temperatures (5–300 K) which span the range of possible surface temperatures in the asteroid belt and outer solar system. Thermal diffusivity and thermal inertia as a function of temperature are calculated from measurements of density, thermal conductivity, and heat capacity. Thermal data can be used to expand databases and improve models of the thermal evolution of planetary bodies.**

Ureilites are a rare type of stony meteorite that has a unique mineralogical composition very different from that of other stony meteorites. Ureilites tend to have a high percentage of carbon (average 3% by weight) in the form of graphite and nanodiamonds. There has not been an identified ureilitic parent body (UPB) and the presence of diamonds is particularly interesting in this regard. Analysis of the carbon inclusions in ureilites can be used to aid in the search for the UPB. We have used a scanning electron microscope (SEM) with an energy dispersive X-ray spectrometer (EDS) in order to identify the mineralogy of NWA 11641. Mineralogical analysis provides further information on the thermodynamic properties of NWA 11641. Identified carbon inclusions will be investigated in further studies.

(1) Introduction:

(a) Measurements of the low-temperature thermodynamic and physical properties of meteorites provide fundamental data for the study and understanding of asteroids and other small bodies. Ureilites are a rare class of stony meteorites with mineralogical compositions that differ significantly from those of other achondrites. They typically contain several weight percent carbon in the form of graphite and diamond.

(b) The origin of diamonds in ureilites remains an open question. Proposed formation mechanisms include (1) static high-pressure growth deep within a large parent body, (2) shock-induced transformation of graphite during impact events, and (3) low-pressure chemical vapor deposition (CVD) processes in the early solar nebula. Distinguishing among these pathways would provide critical insight into the size, structure, and history of a ureilitic parent body, which has not yet been definitively identified.

(c) Thermophysical data are essential for improving models of heat flow and thermal evolution in small bodies. Complementary mineralogical characterization, including microscopic and spectroscopic identification of carbon inclusions, helps constrain the formation conditions of graphite and diamond phases. Together, these approaches work to understand the thermal properties and chemistry of ureilites in order to refine models of astronomical bodies and identify the UPB.

(2) Specific Heat Capacity (c_p) and Thermal Conductivity (κ) vs. T

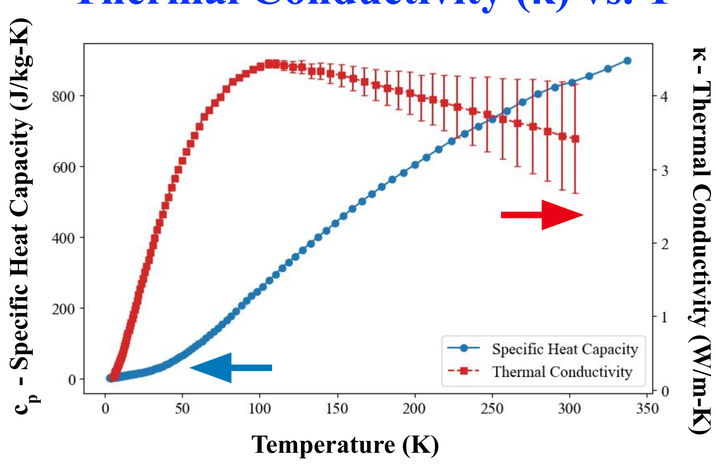
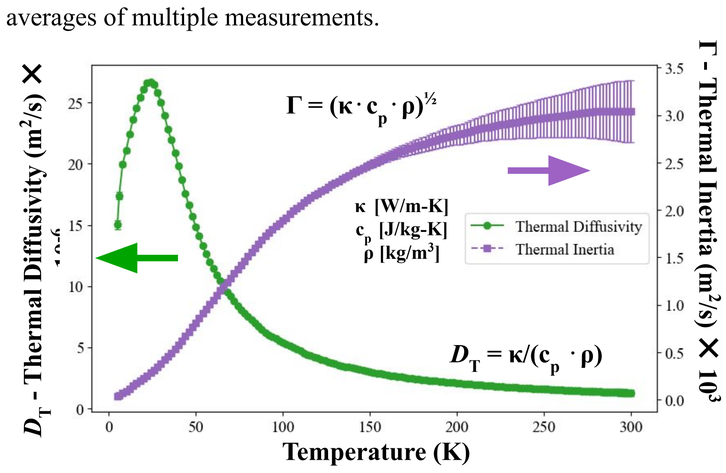


Fig. 1: Specific heat capacity vs. temperature and thermal conductivity vs. temperature measurements are shown. c_p data was measured using a thermal relaxation technique with a sample of ~20 mg and approximate dimensions of ~1 × 3 × 3 mm. κ data was measured with a cooling rate of 0.5 K min^{-1}. Sample dimensions were ~2 × 3 × 10 mm. Complete measurement techniques are described in Ref. [5].

(3) Thermal Diffusivity (D_T) and Thermal Inertia (Γ) vs. T

Fig. 2: Thermal diffusivity vs. temperature and thermal inertia vs. temperature measurements are shown. Thermal diffusivity is calculated from the relationship $D_T = \kappa/(c_p \cdot \rho)$ where ρ is the mass density of the sample. Thermal inertia is calculated from the relationship $\Gamma = (\kappa \cdot c_p \cdot \rho)^{1/2}$. In both calculations, κ and c_p were interpolated from the data in Fig. 1. Mass density ρ was calculated from the relationship ρ = m/V, where m is the mass and V is the volume of the sample. Volume was measured using a blue-light scanner and the sample mass was measured using a mass balance. Values used in calculations are averages of multiple measurements.



(4) SEM and Optical Images

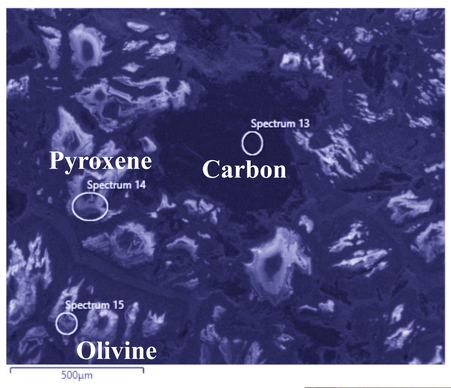


Fig. 3: Back-scattered electron (BSE) image of NWA 11641 showing typical Carbon inclusion ~500 µm in size embedded in silicate matrix of larger Olivine and Pyroxene crystals. Image was acquired using a Scanning Electron Microscope (SEM)

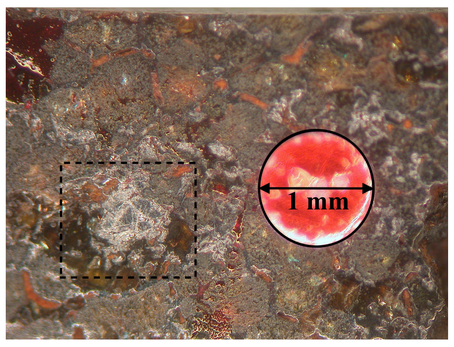


Fig. 4: Image of NWA sample used in SEM and EDX measurements. Approximate area measured by SEM and EDX is outlined. A 1 mm diameter cylinder was added for reference.

(5) EDS

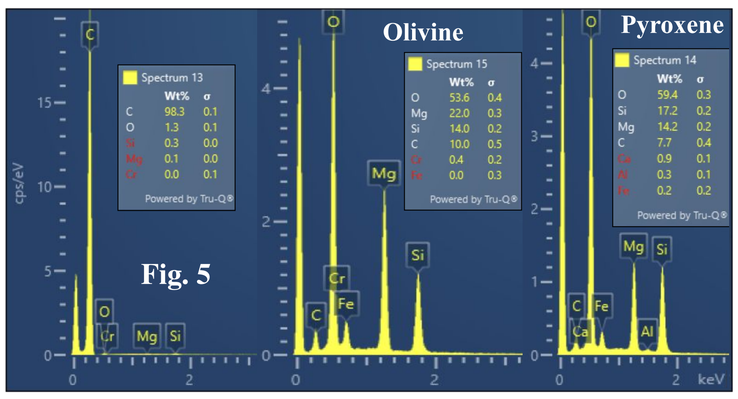


Fig 5: Elemental composition of NWA 11641 carried out using an energy dispersive X-ray spectrometer (EDS) on the areas noted in Fig. 3. This data was used to determine the identity of the silicate matrix. Olivine has the chemical formula X_2SiO_4, where X is a cation, most commonly Fe or Mg. Pyroxene has the chemical formula $XYSi_2O_6$, where X and Y are cations, most commonly Fe, Mg, or Ca.

(6) Conclusions/Analysis:

(a) Our thermophysical data of expand the limited dataset of low-temperature measurements of meteorites. Since the temperature dependence of thermal diffusivity has a considerable effect on thermal modeling of meteorites, significant increases in thermal diffusivity at T < 100 K shown here should add to the knowledge and understanding of meteorite evolution and modeling.

(b) Mineralogical analysis using SEM and EDS confirms the heterogeneous silicate matrix of olivine and pyroxene, along with abundant carbon-rich inclusions. Carbon inclusions we identified will be studied using Raman spectroscopy in order to investigate the thermal history of the ureilites.

References:

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